carbon_lyr.bsn

Purpose

`carbon_lyr.bsn` holds the carbon coefficients that differ between soil layer groups. It populates `carbdb(layer_id)` (rate constants for the litter, microbial, slow, and passive pools) and `org_allo(layer_id)` (the CO2 fractions produced by each pool transformation). The two layer groups currently in use are:

- `layer 1` (surface litter and topsoil)
- `layer 2` (all subsurface layers)

The reader indexes directly into the `carbdb(:)` and `org_allo(:)` arrays. Both are sized `dimension(2)` in `carbon_module.f90` today; a future expansion to additional layer groups requires only a larger array allocation, not a reader change.

Source

- Reader: same routine as `carbon.bsn`, `src/carbon_bsn_read.f90`. The per-layer block is parsed in the second half of the subroutine.
- Target types: `carbon_inputs` (`carbdb`) and `organic_allocations` (`org_allo`) in `src/carbon_module.f90`.
- Filename: hardcoded in the reader as `carbon_lyr.bsn`. **Not listed in `file.cio`** — it rides at a fixed name beside `carbon.bsn`.

Activation

Read only when `bsn_cc%cswat == 2` (the dynamic CENTURY/SWAT-C model; the activation value moved from 1 to 2). Like `carbon.bsn`, the file is **required**: an absent file, an unparseable row, or fewer data rows than the size of `carbdb` (currently 2) triggers `error stop` with a clear message.

Format

- Line 1: title (skipped).
- Line 2: column header (skipped).
- Lines 3..: one row per layer group. Each row begins with an integer `layer` id (1 = top, 2 = subsurface) followed by 11 real coefficients. The reader loops until end-of-file. Rows with an out-of-range `layer` id (less than 1 or greater than `size(carbdb)`) are skipped with a warning to unit 9001 and do not count toward the required row count.

#	Field	Type	Target	Units	Description
1	`layer`	int	(array index)	none	layer group id; 1 = top (surface litter + topsoil), 2 = all subsurface layers
2	`hp_rate`	real	`carbdb(layer)%hp_rate`	1/day	passive humus transformation rate under optimal conditions
3	`hs_rate`	real	`carbdb(layer)%hs_rate`	1/day	slow humus transformation rate under optimal conditions
4	`microb_rate`	real	`carbdb(layer)%microb_rate`	1/day	microbial biomass transformation rate under optimal conditions
5	`meta_rate`	real	`carbdb(layer)%meta_rate`	1/day	metabolic litter transformation rate under optimal conditions
6	`str_rate`	real	`carbdb(layer)%str_rate`	1/day	structural litter transformation rate under optimal conditions
7	`microb_top_rate`	real	`carbdb(layer)%microb_top_rate`	1/day	top-layer microbial activity adjustment
8	`hs_hp`	real	`carbdb(layer)%hs_hp`	none	slow-to-passive humus allocation coefficient
9	`a1co2`	real	`org_allo(layer)%a1co2`	fraction	decomposed metabolic litter routed to CO2
10	`asco2`	real	`org_allo(layer)%asco2`	fraction	decomposed slow humus routed to CO2
11	`apco2`	real	`org_allo(layer)%apco2`	fraction	decomposed passive humus routed to CO2
12	`abco2`	real	`org_allo(layer)%abco2`	fraction	decomposed microbial biomass routed to CO2

Example

`refdata/Lafayette/carbon_lyr.bsn`:

```
carbon_lyr.bsn: layer-keyed carbon coefficients (one row per layer group; 1 = top, 2 = subsurface; extend
   layer       hp_rate       hs_rate   microb_rate   meta_rate   str_rate   microb_top_rate   hs_hp   a1co
       1      1.2e-05      2.92e-04        0.0164      0.0405     0.0107            0.0164    0.05    0.60
       2      1.2e-05      1.81e-04          0.02      0.0507     0.0134              0.02    0.05    0.55
```

Calibration

All 11 per-layer fields are wired in `cal_parm_select.f90`. The calibration target is the `carbdb(ly)` or `org_allo(ly)` element selected by the `ly` index in the calibration record (see Parameter change files):

- `hp_rate`, `hs_rate`, `microb_rate`, `meta_rate`, `str_rate`, `microb_top_rate`, `hs_hp` target `carbdb(ly)%*`.
- `a1co2`, `asco2`, `apco2`, `abco2` target `org_allo(ly)%*`.

Each `case` branch in `cal_parm_select.f90` guards `ly` to the range `[1, size(carbdb)]` (or `[1, size(org_allo)]`), so out-of-range layer indices in a calibration row are silently no-op rather than crashing.

Related files

- `carbon.bsn`. Basin-wide scalar carbon parameters.
- `codes.bsn`. The `cswat` switch.
- Soil carbon (model theory). Pool definitions and Century-style transformation equations.